APPLE'S ON-DEVICE FOUNDATION MODEL, REBUILT: A COMPLETE FROM-WEIGHTS RECONSTRUCTION OF `AFMPLUS-V11.2-IFP` (AFM 3 CORE ADVANCED) AND A PORTABLE LLAMA.CPP RUNTIME

YU YAO-HSING

Abstract. We reconstruct Apple's current on-device foundation language model—asset `com.apple.fm.language.instruct_3b.base.generic_sparse`, display version `afmplus-v11.2-ifp`, which Apple calls *AFM 3 Core Advanced*—entirely from the files Apple Intelligence installs on a Mac, and run it in llama.cpp on any operating system. The model is a 56-layer, 20.3-billion-parameter transformer with a 1536-wide residual stream, grouped-query attention (16 query and 8 key/value heads), KV-cache sharing across its last 21 layers, a post-normalised ("x/y") residual with four learned RMSNorm gains per layer, and an Instruction-Following-Pruned sparse feed-forward network: 23 always-on experts plus 23 of 348 routed experts, selected by a separate router network evaluated at layer 12. Every 2-bit weight is carried into GGUF *bit-exactly* (55 million spot-checked weights, zero differences), and the tokenizer reproduces Apple's token counts on 81/81 test texts. The prompt template and the expert-selection schedule, which the model files do not contain, are recovered from Apple's inference frameworks and confirmed by Apple's public token counter and runtime log. Nothing in the reconstruction is trained or fitted.

Against Apple's live model at temperature zero, the reconstruction's next-token choice agrees with Apple's on **97.0%** of answer tokens (95.3% on a harder 230-prompt set), and 17 of 40 free-running answers are identical to Apple's character for character. The residual disagreements are near-ties whose resolution depends on the exact arithmetic of Apple's compiled Neural Engine program. We also reconstruct the models that ship alongside it and validate each against Apple's own wherever a reference exists: the 26 Apple Intelligence feature adapters (rank-48 LoRAs; the translation and notification-urgency adapters reproduce Apple's outputs); the 9M-parameter speculative-decoding draft model, which reproduces 99.2% of the clear accept/reject decisions logged by Apple's runtime and makes speculation in llama.cpp lossless; the 358M-parameter image encoder, which matches Apple's encoder at cosine 0.98–0.99 per token and yields captions identical to Apple's for 6 of 13 images; and the 300M base model (`afmplus-v11.0-pico`) with its adapters, among them the safety guard, which predicts the public API's guardrail decision on 95 of 100 prompts, and its own image encoder (cosine 0.99 per token on photos).

For much of this work the assembled model produced no meaningful predictions. The cause was a single misaligned read: every normalisation gain was taken one element early. The defect was located with probes *calibrated* on a healthy Apple model—Xcode's on-device code model, whose MPSGraph package we dumped to readable MLIR, executed on the GPU from our own process and reproduced exactly in NumPy. The report documents the model's full structure, the storage formats, the debugging instruments and their calibration, the prompt template and expert schedule recovered from Apple's runtime, the llama.cpp implementation, a standalone converter, the negative results, and an analysis of what still separates the reconstruction from exact agreement.

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1. Summary of results

Model	`afmplus-v11.2-ifp` = AFM 3 Core Advanced (on-device, sparse IFP)
Size	20.30 B parameters stored; 1–4 B active per prompt (ifp1/2/3 budgets)
Layers	56 = 12 dense + 23 sparse (own K/V) + 21 sparse (reuse layer-34 K/V)
Width	$d = 1536$; 16 Q / 8 KV heads × 128; dense FFN 3072; expert width 256
Weights	2-bit palette $\{-1.5, -0.5, 0.5, 1.5\}$ × fp16 row scale; int4 embedding
GGUF	6.31 GiB, Q2_K (bit-exact) + Q4_0 (exact); custom arch `afm`
Fidelity	97.0% teacher-forced top-1 vs Apple (40 prompts, 1127 tokens), 95.3% on 230 prompts; 17/40 answers identical
Runtime rules	`afm_cider_v2` template and IFP expert schedule, decoded from Apple's frameworks
Tokenizer	matches Apple's `SystemLanguageModel.tokenCount` on 81/81 texts
Runtime	llama.cpp patch: Metal, CUDA, Vulkan, CPU; about 150 tokens/s prompt processing, 12–21 tokens/s generation (M4)
Converter	`afm2gguf.py`: 2–3 min, reads only the local assets, self-validating
Draft model	Apple's speculative-decoding draft (`v11-ifp-9m`, 12 layers, $d = 256$) decoded: reproduces the accept/reject decisions of Apple's own draft, which Apple's runtime logs (95.6% of 1331, 99.2% of the 1134 clear ones); 43% next-token agreement with the 3B model; exact `afm-draft` GGUF; speculation lossless (40/40) but not faster in llama.cpp (Section 11)
Image encoder	24-layer windowed ViT + 3 × 3 merger (358 M) decoded; matches Apple's own encoder (cosine 0.98–0.99 per token on photos); captions 94.0% teacher-forced (Apple's own encoder: 92.9%), 6/13 identical; llama.cpp projector `afm` (Section 12)
300M base	`afmplus-v11.0-pico`, the base of Apple's 300M classifiers (24 layers, $d = 1024$) decoded: perplexity 21–31 on plain English with its default adapter, 78% induction copying, coherent chat answers; `afm-draft` GGUF (Section 13)
Safety guard	Apple's 300M safety classifier and 11 other task adapters decoded as LoRAs; the guard answers `safe` for benign text, labels harmful text in its own `category,action/mention` format, and with Apple's policy table predicts the public API's decision on 95 of 100 prompts (Section 13)
300M images	the 300M model's own image encoder (same structure as the 3B's, output 1024) decoded; matches Apple's own encoder at cosine 0.99 per token on photos; its people counter is right on 10 of 10 photos with 0–2 people, and its image safety classifiers pass every benign image (Section 13)
Principal defect corrected	norm-gain table read 2 bytes (one fp16 element) early (Section 6)

Table 1. Principal results. All measurements in this report were taken on an M4 Mac running macOS 27.

Code: `https://github.com/yu314-coder/afm-ifp-teardown` (llama.cpp patch, converter, build workflow). No Apple model data is published; the converter rebuilds the GGUF from the assets on the user's own machine.

2. Which model is this?

Apple's third-generation foundation models (announced June 2026) comprise two on-device language models and several Private Cloud Compute models (Table 2). The macOS 27 SDK exposes the on-device model as `SystemLanguageModel` and the server models through `PrivateCloudComputeLanguageModel`. The assets on the device contain exactly one foundation language backbone, `afmplus-v11.2-ifp`; the server models are present only as adapter metadata (`afmplus-v8.0-150b`, `afmplus-v9.0-3b-pcc`).

Name	Where	Size	Architecture	On this Mac?
AFM 3 Core	device	3 B	dense	no separate asset found
AFM 3 Core Advanced	device	**20 B**, 1–4 B active	**sparse, IFP**	**yes: `afmplus-v11.2-ifp`**
AFM 3 Cloud	PCC	n/a	n/a	metadata only
AFM 3 Cloud Pro	PCC (GPUs)	n/a	n/a	no

Table 2. Apple's 2026 lineup as described in public WWDC 2026 coverage (secondary sources; Apple's own documentation does not list sizes). The match for v11.2 rests on our measurements: llama.cpp counts 20.30 B parameters in the converted model; its feed-forward network is Instruction-Following-Pruned; and the Foundation Models API ships three compute budgets (`fm_api_generic_1b/2b/3b`) that select 5, 14 or 23 routed experts—the "1–4 B active per prompt".

The reconstructed model is therefore *AFM 3 Core Advanced*. It is not a "Pro" model: the Pro model runs only on Apple's servers and is not present on the device.

3. Model assets

Apple Intelligence downloads its models as MobileAssets under `/System/Library/AssetsV2/com_apple_MobileAsset_UAF_FM_GenerativeModels`. The model files sit in an APFS cryptex that the system grafts into the asset's `.AssetData` directory.

File	What it holds
`binary_0.hwx` (694 MB) in `main-h16g-delegates/MPSGraph/`	The compiled ANE program: 108 procedures (prompt / extend / dense-only / sparse-only / readout, per sequence length and IFP budget). Segment `__KERN_3` holds every attention and dense-FFN weight; `__TEXT,__const` holds the norm gains; the task stream holds the int8 K/V calibration.
`main-h16g.odix` (263 MB)	The orchestration program (ODIE IR). Holds the int4 token embedding (text, audio) and its per-token fp16 scales, and the RoPE tables.
`ifp/ifp_rasterized_weights.bin` (4.9 GB)	All 44×380 experts: 2-bit codes, gate/up and down row scales.
`ifp/project_experts.mlasset`	The IFP router network (fp16).
`ifp/config.json`	IFP budgets: ifp1/2/3 = 5/14/23 active experts; `expert_selection_frequency` = 32.
`tokenizer` (separate asset)	328 192-entry SentencePiece-BPE vocabulary in a flat Apple format (262 144 text tokens + audio/voice tokens).

Table 3. Files used by the reconstruction.

Reading Apple's graphs. A private method on `MPSGraphExecutable`, `-getIR`, returns a loaded package's graph as MLIR text. On AFM it yields only placement operations and 108 calls into the ANE program, but with exact entry signatures (for example, the two 64×128 RoPE tables arrive as runtime inputs). On Xcode's code-completion model (CodeLM v3, a GPU model built with the same modelling library, *tamm*) it yields the complete source-level graph. CodeLM serves as our healthy reference model (Section 7).

4. Architecture

Figure 1 shows the model as a whole. The subsections that follow give its dimensions, the structure of a layer, the attention mechanism and the sparse feed-forward network.

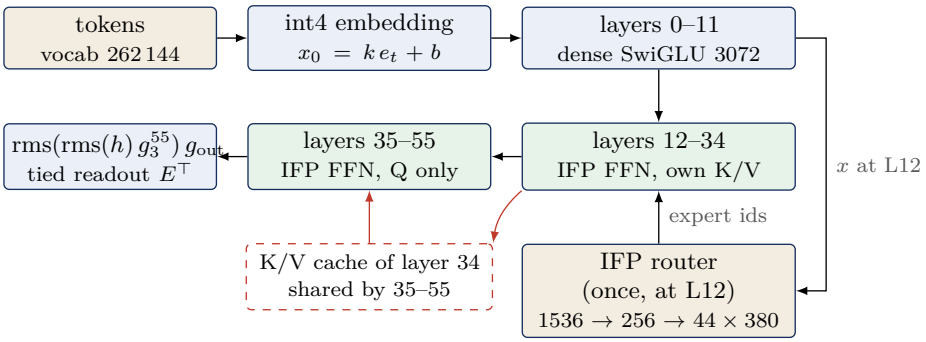


Figure 1. The whole model. The router reads the normalised input of layer 12 and emits routing logits for all 44 sparse layers at once. Layers 35–55 compute queries only and attend to the K/V cache written by layer 34.

Quantity	Value	Evidence
layers	56 (12 dense, 23 sparse own-KV, 21 sparse KV-reuse)	chunk-set plan; task stream
hidden / head dim	1536 / 128	conv shapes
heads	16 query, 8 key/value; query head h uses KV head $\lfloor h/2 \rfloor$	attention task addressing; same-token probe
dense FFN	SwiGLU, width 3072 (layers 0–11)	conv shapes; SiLU lookup table on the gate
experts	380 per sparse layer, width 256; 0–22 always on; 32–379 routable	router mask; static kernels = experts 0–22
RoPE	NEOX half-split, $\theta = 5 \times 10^5$, 128 dims	odix cos table (row 0 = 1, halves equal)
norm ε	1.0014×10^{-5}	ANE register; identical in CodeLM
attention scale	$1/\sqrt{128}$	post-scale = $(1/s_K)/\sqrt{128}$ in the hardware; CodeLM `sdpa`
vocabulary	262 144 text tokens (Gemma-3 family layout), BOS 2, end-of-turn 106	tokenizer asset
context	8192	asset metadata

Table 4. Architecture parameters and where each was read.

4.1. Dimensions.

4.2. **The layer: a post-normalised "x/y" residual.** Each layer carries four learned RMSNorm gains $g_0, \ldots, g_3 \in \mathbb{R}^{1536}$. Unlike a pre-norm transformer, both residual additions add *normalised*

values:

$$
\begin{aligned}
x_L &= \mathrm{rms}(h_{L-1}) \odot g_3^{L-1} \qquad \text{(layer 0: } x_0 = k\, e_t + b,\ \text{no norm)} \\
h'_L &= x_L + \mathrm{rms}\big(\mathrm{Attn}_L(x_L)\big) \odot g_0^L \\
y_L &= \mathrm{rms}(h'_L) \odot g_1^L \\
h_L &= y_L + \mathrm{rms}\big(\mathrm{FFN}_L(y_L)\big) \odot g_2^L
\end{aligned}
$$

with $\mathrm{rms}(v) = v/\sqrt{\mathrm{mean}(v^2) + \varepsilon}$, gains used literally (no $1 + g$), and a final output $\mathrm{rms}(\mathrm{rms}(h_{55}) \odot g_3^{55}) \odot g_{\text{out}}$ multiplied by the tied embedding. The residual wiring was read from the DMA pattern of the ANE program: the normalised attention input is written to DRAM scratch and read back by the attention residual add; the FFN residual add reads the normalised FFN input. Only this wiring reproduces Apple's outputs (standard residual: 0% top-1).

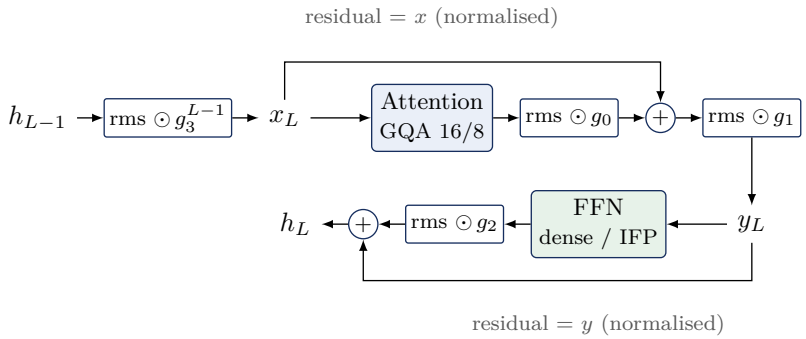


Figure 2. One layer. Both residual additions add normalised streams; g_0 and g_2 are zero-centred (about 30% negative entries), g_1 and g_3 small and positive. Zero-centred literal gains are normal in Apple's modelling library: a healthy Apple model (CodeLM) shows the same pattern.

4.3. **Attention.** Attention is standard grouped-query attention with two Apple-specific details: the query/key normalisation sits after RoPE, and the key/value cache is int8 (Figure 3).

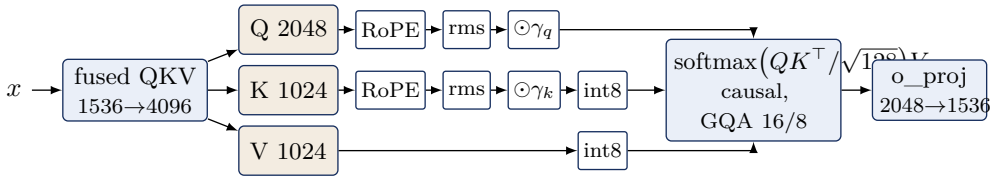


Figure 3. Attention as executed on the ANE. RoPE comes before the per-head RMS norm and its 128-entry gain (the same order appears in CodeLM's source graph). Keys and values are stored in an int8 cache with per-layer calibrated scales; layer 34 has no key gain. Layers 35–55 compute only Q and read layer 34's cache.

4.4. **The Instruction-Following-Pruned feed-forward network.** Sparse layers add two SwiGLU blocks without any gating weights:

$$
\mathrm{FFN}(y) = \underbrace{\sum_{e=0}^{22} \mathrm{SwiGLU}_e(y)}_{\text{always on}} + \underbrace{\sum_{e \in \mathcal{S}(y)} \mathrm{SwiGLU}_e(y)}_{|\mathcal{S}|=23 \text{ routed}}, \qquad \mathrm{SwiGLU}_e(y) = W_e^{\text{down}}\big(\mathrm{SiLU}(W_e^{\text{gate}} y) \odot W_e^{\text{up}} y\big),
$$

each expert being 256 wide. The ANE program runs the always-on experts from static kernels and the routed experts from a mutable kernel patched at runtime from `ifp_rasterized_weights.bin`. Both halves matter: the always-on experts alone reach 62.9% top-1, the routed experts alone 1.4%, both together 87.5% (measured with the single-turn template).

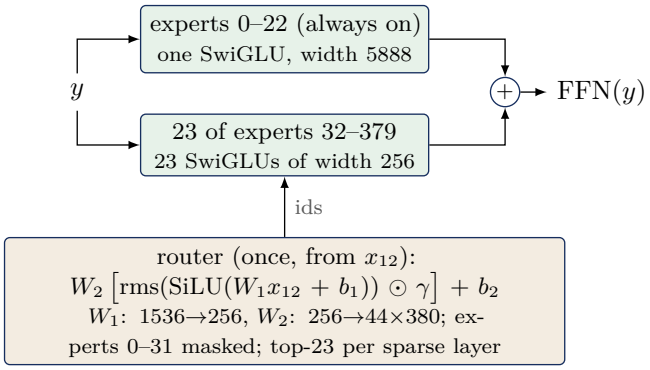


Figure 4. The IFP feed-forward network of a sparse layer and the router that feeds it. The sum is unweighted: the router only selects.

When experts are chosen. Apple's runtime does not route per token. It runs the dense layers over the whole prompt, takes the layer-12 input of the *last* prompt token, runs the router once, and uses that expert set for every prompt token and for generation; it re-picks from the current token only at absolute positions that are multiples of 32. Section 8 shows how this was read out of Apple's inference framework.

5. How the weights are stored

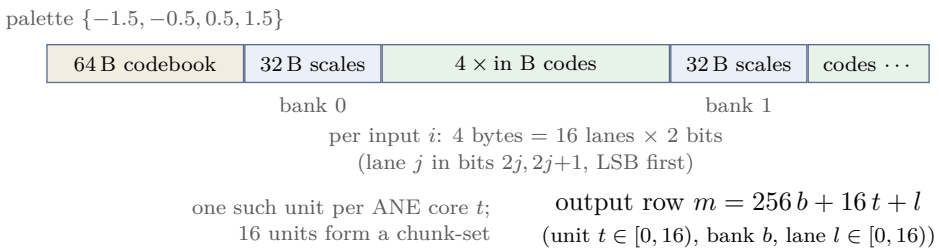


Figure 5. The 2-bit weight unit. A projection is split across the 16 ANE units bank-cyclically; a chunk with more banks holds the lower bank slots. The value of a weight is $w = s_m\,(c - 1.5)$ with $c \in \{0, 1, 2, 3\}$ and s_m the fp16 row scale. Experts in `ifp_rasterized_weights.bin` use the same codes with a simpler order: code $p \mapsto$ (row $16\lfloor p/(16\,\mathrm{in})\rfloor + p \bmod 16$, input $\lfloor p/16\rfloor \bmod \mathrm{in}$).

Exact GGUF encoding. llama.cpp's Q2_K stores, per 256 weights, sixteen 4-bit sub-block scales sc and mins m, a super-scale d and super-min $d_{\min}$, and evaluates $w = d\,sc\,c - d_{\min}\,m$. Choosing $sc = 2$, $m = 3$ and $d = d_{\min} = s/2$ gives

$$w = \tfrac{s}{2} \cdot 2c - \tfrac{s}{2} \cdot 3 = s\,(c - 1.5),$$

and halving an fp16 number is exact (barring underflow, which never occurs here). The int4 embedding ($w = s_t\,q$, $q \in [-7, 7]$, per-token fp16 scale s_t) maps exactly onto Q4_0 with $d = s_t$ and nibble $q + 8$. Dequantising the GGUF with gguf-py reproduces Apple's weights with maximum absolute difference 0.

Layout facts and how they were ground-truthed. We compiled known weights through Apple's own ANE compiler (`mil_to_hwx -a h16`) and decoded the result: the output axis is the bank-cyclic stripe above, the input axis is natural, the codebook and row scales are as in Figure 5, and constants (gains, biases) are stored in natural order. A calibrated copy-circuit probe (Section 7) independently confirms the stripe, the head order and the V/O pairing: every alternative layout scores at chance.

6. The norm-gain table and its alignment

All gains live in one contiguous fp16 table inside `__TEXT,__const` of `binary_0.hwx`:

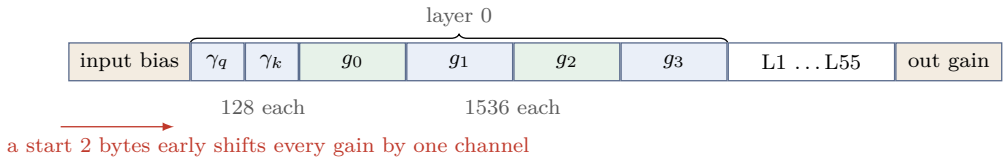


Figure 6. The gain table. Layers 0–33 carry γ_q and γ_k; layers 34–55 only γ_q. Every vector length is a multiple of 64 bytes, so a correct start is 64-byte aligned.

Finding 1. An initial parse located the table by statistical signatures and accepted a start two bytes early. Every RMSNorm therefore multiplied channel r by the gain of channel $r-1$. Because g_0 and g_2 are zero-centred, this flipped the sign of about a third of every residual write at every layer, and the assembled model predicted nothing (0% top-1, no induction).

How it was found. No tensor was defective in isolation: each was correctly laid out and correctly paired. The decisive step was to calibrate probes on a healthy Apple model and then to compare *reader–reader* with *reader–writer* relations. In AFM, query–key and gate–up relations were as healthy as in CodeLM, but value–output (copy heads) and up–down (FFN neuron coupling) relations were weak. In a trained network, the sign of the coupling between up-row i and down-column i agrees with the sign of the gain that scales the write. In AFM the *number* of negatively coupled channels equalled the number of negative entries of g_2 in every layer—while their *positions* disagreed (Figure 7, left). Matching each channel's 11-layer sign signature against the table gave a constant offset of one element. Read aligned, the agreement rises to 98–99.9% (right) and the model works: 66% top-1 against Apple's outputs immediately, rising to 87.5% with the llama.cpp implementation and exact tokenization.

7. Validation instruments and their calibration

The oracle. `afm -t 0` (a small Swift wrapper around the Foundation Models API) produces Apple's greedy answers; at temperature zero, Apple's top-1 token at every answer position is the next token, so teacher-forced top-1 agreement is an absolute measure. We use 40 prompts (1127 answer tokens). The API exposes no probabilities.

A healthy reference: CodeLM v3. Xcode's on-device code model ships as a GPU MPSGraph package. Its MLIR (via `-getIR`) is a complete, readable tamm transformer—pre-norm, standard residual, RoPE then QK-norm, SwiGLU, KV reuse from layer 34, int4 weights with group-32 scales—with the same backbone shape as Apple's other 3B-class models (56 layers, 35 KV-owning, 2048 wide, 16/2 heads). We ran it on the GPU from our own process through `MPSGraphExecutable` (100% induction accuracy) and reproduced its logits in NumPy exactly (correlation 1.00000, 100% argmax agreement). Its tokenizer and embeddings are unrelated to v7-3b's, so it cannot donate weights—but it calibrates instruments.

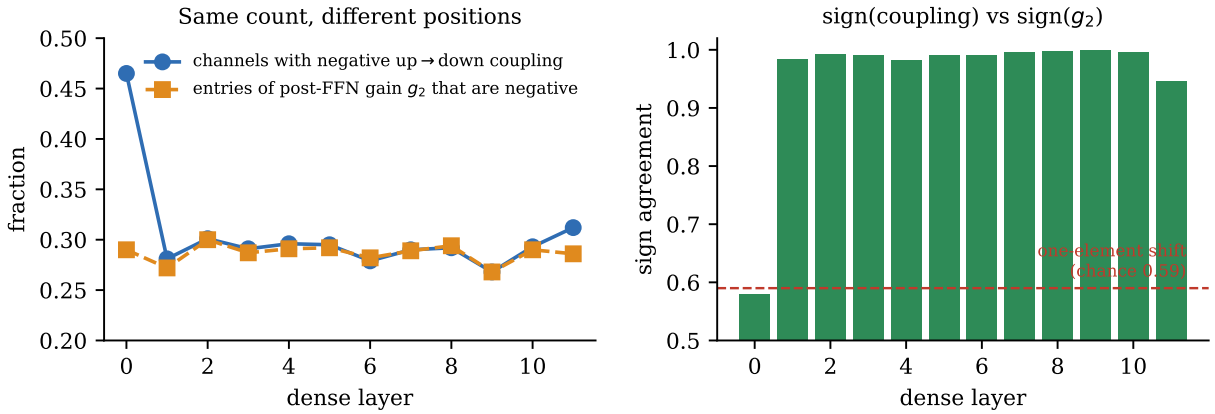


Figure 7. Left: per layer, the fraction of residual channels with negative up→down coupling (from the weights) and the fraction of negative entries in the decoded post-FFN gain g_2—equal counts. Right: per-channel sign agreement after reading the table aligned; a one-element shift gives chance level. Layer 0 is the exception because its input is the raw embedding.

What the calibration showed.

- *Zero-centred literal gains are normal* in tamm (CodeLM's pre-attention gain: mean +0.003, 46% negative).
- *The embedding–weight "alignment ratio" used earlier in this work is not diagnostic*: a healthy model scores the same (1.0–1.4).
- *The OV copy probe is diagnostic*: CodeLM has several heads that map a token to itself at rank 0–16, even after naive 2-bit rounding; decoded AFM had none below rank 1000 (Figure 8). Every alternative V/O layout scored at chance, which ruled out layout errors and pointed at the gains.
- *The same-token QK probe* found AFM's query–key circuits healthy (rank-0 heads) and confirmed the head pairing $h \mapsto \lfloor h/2 \rfloor$.

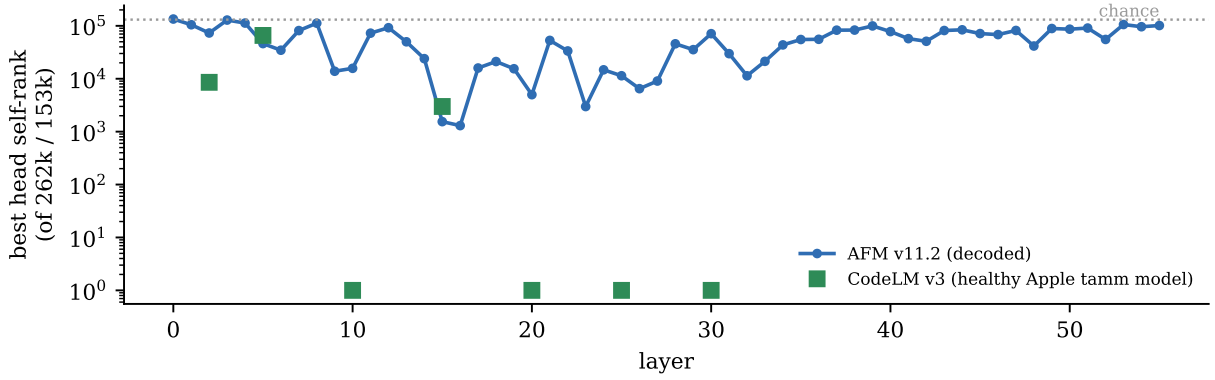


Figure 8. Copy-circuit probe: for each layer, the best attention head's median rank of the input token in $E\,W_O W_V\,e_t$. Healthy CodeLM has rank-0 heads; AFM's raw $W_O W_V$ peaks at weak copying because in AFM the write passes through the signed gain g_0, whose positions were scrambled by the table shift.

Apple's int8 calibration as a ground truth. The ANE program stores per-layer scales for its int8 K/V cache. Apple chose them from real activations, so they bound the true magnitudes. With the corrected gains, the reconstruction's values stay inside Apple's ranges, and its keys stay inside up to the 99.9th percentile; only the most extreme 0.1% of keys would clip (Figure 9). This is an independent check of the internal scales. (Figure 9 was produced before the layer-0 corrections of Section 14; with the correct layer-0 input the layer-0 values use only part of Apple's range, so Apple's layer-0 scale is not a tight bound.)

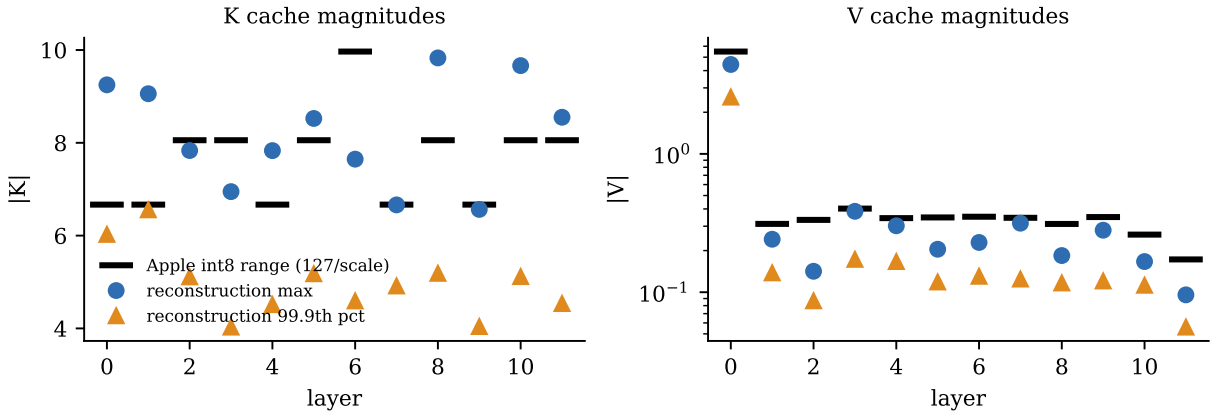


Figure 9. Reconstructed key and value magnitudes (dense layers, one prompt) against Apple's calibrated int8 ranges. Layer 0's values are large because its input is the un-normalised embedding.

8. Apple's runtime: the prompt template and the expert schedule

Two things the model files do not contain decide what the model sees and which experts it uses: the prompt template (the metadata only names it, `afm_cider_v2`) and the rule for choosing experts. Both live in code, in two private frameworks of the dyld shared cache: `TokenGenerationCore` renders and tokenizes prompts, `TokenGenerationInference` drives the model. `dyld_info -disassemble` prints both with Swift symbol names. One detail matters for reading them: Swift stores string literals of up to 15 bytes inside the instruction stream (`mov`/`movk` immediates), so `<start_of_turn>` and the other markers do not appear in a strings dump.

8.1. The prompt template.

`PromptPreprocessingTemplateAfmCiderV2Convention` takes its special-token table from the older Cider v1 convention and overrides only audio and reasoning entries. Decoding the immediates gives turn start `<start_of_turn>`, turn end `<end_of_turn>`, and roles `system\n`, `developer\n`, `user\n`, `model\n`; the "thought budget: none" and "developer static instructions" entries are empty. Its `cacheableSystemPrefix` concatenates turnStart, the system role with Apple's default instructions, the thought budget, turnEnd, a newline, then turnStart and the developer role. A FoundationModels conversation therefore renders as

<bos><start_of_turn>system\n*default*<end_of_turn>\n
<start_of_turn>developer\n*instructions*\n<end_of_turn>\n
<start_of_turn>user\n*prompt*<end_of_turn>\n<start_of_turn>model\n

where *default* is the `systemInstructions.default` text of the adapter metadata ("A conversation between a user and a helpful assistant. You are a foundation model developed by Apple. . . .") and the developer turn holds the app's instructions. The public API verifies every detail: `SystemLanguageModel.tokenCount(for:)` accepts transcript entries and returns the length of the rendered prompt, which equals the length the inference service logs ("asked to run 71 tokens") for

the same conversation. Table 5 compares it with our rendering. A template that folds everything into one system turn and omits the instructions passed by the reference client scores markedly lower; with the exact template, teacher-forced agreement rose from 87.5% to 90.5% and the first answer token from 30 to 38 of 40.

Transcript (I = instructions, P = prompt, R = response)	Apple `tokenCount`	our template
P only	16	16
I (empty), P	62	62
I only	55	55
I, P	71	71
I, P, R, P	86	86

Table 5. Prompt lengths. A session without instructions still carries the system turn and an empty developer turn (62 tokens, seen in the live log); each instruction segment ends with a newline; earlier answers are closed with `<end_of_turn>\n`.

8.2. **The expert schedule.** The inference log of a single request already shows the shape of the algorithm: a "prompt expert compute function" (`lora_48_ifp3_r48_prompt_opt_dense_only_256_64`, the dense layers only) runs over all prompt tokens, then "Picking IFP experts at position=70 isPromptPick=true" for a 71-token prompt, then the sparse layers run over the *whole* prompt again with the picked experts. The dense-only function returns a `1536×64` fp16 tensor per 64-token chunk, one column per token. In `TokenGenerationInference`, `computePromptExpertState` builds the router input with `IFPDenseLayerOutput.makeExpertsTensor(tokenIndex:)`, whose loop copies the 1536 values of a single column into a 1536×1 tensor—no mean, no accumulation. The router (`project_experts`, the network of Figure 4) then yields the top-23 experts per sparse layer. During generation, `repickExpertsAtBoundary` calls the same picker, and `IFPExpertPicker` rejects any intermediate pick whose position is not a multiple of `expert_selection_frequency` (an `sdiv`/`msub` remainder test against 32). With prompt length P, token t therefore uses the experts chosen from x_{12} at

$$s(t) = \max(P - 1,\ 32\lfloor t/32 \rfloor),$$

so every prompt token uses the last prompt token's choice. Averaging over the prompt instead is clearly wrong (79.3% teacher-forced), and the single-token rule outperforms per-token routing (Section 14).

An outside check of the router. Each pick also logs `CacheExpertExpertCache: ... replaced=`n, the number of (layer, expert) choices that changed. We ran six long generations on Apple's model, collected its 47 re-picks, replayed the same token sequences through our model and counted how many of our choices change at the same positions. With the single-token rule on the layer-12 attention input the two series agree closely (Figure 10: mean 26.7 versus 27.1 experts, correlation 0.84); prefix means correlate at 0.21–0.53 and the FFN-input variant changes three times too many. This validates the router input, the router network and the rule independently of any answer text.

9. The llama.cpp implementation

The patch adds one architecture, `afm`, in `src/models/afm.cpp`, together with the supporting changes listed below.

- **Graph.** Layer 0 takes the dequantised embedding row as is (no multiplier) and, for generated tokens other than re-pick tokens, adds the input bias (Section 14); each later layer normalises the previous output with g_3; queries and keys use `rms_norm`, NEOX RoPE, then a gain multiply; the residuals follow Figure 2. Sparse layers run the always-on experts as a "shared

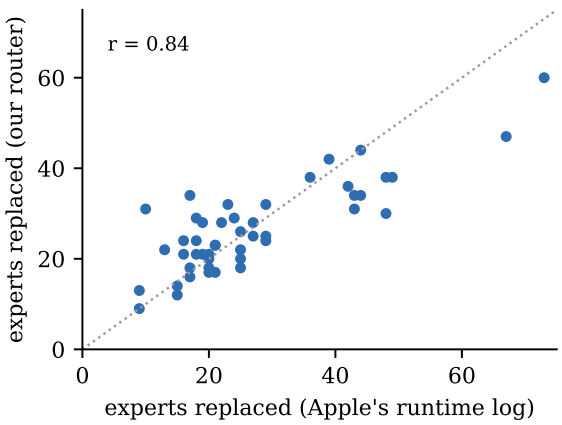


Figure 10. Experts replaced at each re-pick during generation: Apple's runtime log versus our router replaying the same tokens (47 re-picks in six generations).

expert" SwiGLU of width 5888 and the routed experts with `mul_mat_id`, summed by one row reduction with no weights. The router network lives on layer 12 and its logits are sliced per layer. The output applies the double norm and the tied embedding.

- **KV reuse.** Layers 35–55 call attention with null K/V, and the cache maps them to layer 34. This needed a two-line fix in the non-sliding-window attention path, which assumed K/V were always present.
- **Metal kernels.** Metal's batched expert matmul is instantiated for specific experts-per-token counts; we added 14 and 23 (ifp2, ifp3). CUDA uses a generic path for such counts, and Vulkan supports Q2_K expert matmuls, so no change is needed there.
- **Apple's expert schedule.** The router's logits for each sequence persist in a small device buffer (64 sequence slots) owned by the model. A custom graph input computes, per micro-batch on the host, which logits every token uses: a run of several tokens of one sequence is a prompt chunk and takes its last token's logits, which are stored; a single decode token re-picks only when its position is a multiple of 32 and otherwise reads the stored choice. Because the indices are data rather than graph structure, llama.cpp's graph reuse and multi-sequence batching stay correct. Speculative verify batches are handled as single decode tokens, with two position-tagged picks per sequence so that rejected draft tokens leave no trace (Section 11). `AFM_ROUTE_TOKEN=1` restores per-token routing.
- **Embedding scale.** llama.cpp's generic `build_inp_embd()` multiplies by the GGUF's `embedding_scale` (a Granite feature), so an architecture graph that applies the scale itself applies it twice. Apple uses no multiplier (Section 14); the `afm` loader therefore ignores the key, which also keeps GGUFs written by earlier versions of the converter correct.
- **The Neural Engine SiLU.** The activation is the ANE's piecewise-linear table, built from standard ggml operations: a clamp, a magic-constant floor, two 38-entry `get_rows` tables and a relu for the right tail, then rounded to fp16. It costs about 45% of prompt speed and 20% of generation speed on Metal; `AFM_EXACT_SILU=1` switches back to the exact function.
- **Router on the CPU.** The router's top-23 margins (median 0.003) are close to the error of Metal's tiled matmul ($8 \cdot 10^{-5}$ against $2 \cdot 10^{-7}$ on the CPU; 0.002 expert swaps per layer-pick). Its weights are therefore placed on the CPU, which makes picks identical across backends. The routing state and SiLU tables stay on the layer's device. Agreement is unchanged within noise.

- **int8 KV emulation.** The GGUF carries Apple's per-layer K/V scales; keys and values are rounded and saturated to int8 before caching. Rounding uses the fp32 magic constant $1.5 \cdot 2^{23}$, because `ggml_round` misbehaved on Metal.

On an M4 the model processes prompts at about 150 tokens/s and generates 12–21 tokens/s. The patch applies to upstream llama.cpp at the base commit recorded in the repository; the repository's workflow builds Windows (CPU, Vulkan, CUDA), Linux (CPU, Vulkan) and macOS (Metal) binaries from it.

10. Apple Intelligence feature adapters

Apple Intelligence features ship as rank-48 LoRA adapters for this backbone: 26 on the test machine, covering summaries, proofreading, mail replies, smart replies, translation, message actions, reminders, Image Playground suggestions, Photos memories, dictation and others. Each asset holds `lora_i`N`_r48_48_constant_data.bin`, where N is the IFP level the feature runs at. Its metadata holds Apple's prompts for the feature (system instructions and content templates, in the `afm_cider_v2` format) and the same backbone signature as the base model. All of them carry the empty-input SHA-256 as their "adapter signature", so that field says nothing about whether an adapter is trained. The FM API's own adapter is untrained (B RMS $3 \cdot 10^{-5}$), while the feature adapters are trained (B RMS $2 \cdot 10^{-3}$ to $3 \cdot 10^{-2}$).

Layout. A file holds 218 (A, B) pairs as fp16: A as $[48, \text{in}]$, then B as $[\text{out}, 48]$. The pairs come in the order of the package manifest, which sorts the tensor names as strings: the dense segment (o, q, k, v, `linear_0`, `linear_1`, down) $\times 12$, then kv-reuse (o, q) $\times 21$, then standard sparse (o, q, k, v) $\times 23$, with the layers inside each segment in the order 0, 10, 11, 1, 2, ... rather than numerically. `linear_0` is the gate and `linear_1` the up projection, and the adapter is applied at scale 0.5. All three points match the 300M adapters (Section 13).

Checked against Apple. Apple's public Translation framework is served on device by the `machine_translation` adapter at ifp1 (the model manager's log names the asset, and the request produces speculative-decoding log lines), and its output is deterministic. The prompt is not in the adapter metadata. We recovered it from the `TranslationInference` framework and confirmed it with the token counts Apple logs for each request (51, 61, 63 and 65 tokens for four probes, all reproduced): the system turn is *"Conversation between a user and a friendly assistant. You are an expert translator."* and the user turn is *"Translate from S to T.\nS phrase:* text *\n\nT phrase:"*, with language names such as "American English" and "Chinese, Traditional".

On 200 translations of 40 English sentences into German, French, Spanish, Japanese and Traditional Chinese, teacher-forced on Apple's output, our adapter picks Apple's token at 3,380 of 3,424 positions (98.7%); the base model at ifp1 without the adapter reaches 89.2%. Table 6 gives the figures by target language; the adapter's contribution is largest for Japanese and Chinese, where the base model alone is weakest. Free-running in llama.cpp, 166 of the 200 translations are identical to Apple's. Apple translates each sentence of a multi-sentence input separately, which our test did not do. The reference is essential. An alternative decode (numeric layer order, scale 1, `linear_0` = up) passes a likelihood test on a small set of self-constructed proofreading pairs, yet scores 0.7% against Apple's translations: a decode of this kind helps slightly at a small scale and diverges at full scale. Writing Tools' Proofread action cannot serve as a reference, because it is served by a Private Cloud Compute model.

Target language	Tokens	Reconstructed adapter	Base model alone
German	725	98.8%	93.0%
French	777	99.5%	94.2%
Spanish	659	99.5%	95.8%
Japanese	659	98.2%	84.4%
Chinese, Traditional	604	97.4%	76.3%
All	3,424	98.7%	89.2%

Table 6. Agreement with Apple's Translation framework by target language: share of Apple's output tokens that the model ranks first, teacher-forced, for 40 English sentences per language, at ifp1.

A second adapter: notification urgency. Apple's suggestion service classifies every app notification with the `urgency_classification` adapter (ifp3) and logs its verdict ("Notification urgency: true"). A small test application that posts local notifications therefore provides a reference for a second adapter at a different IFP level. The adapter is a one-token classifier: the notification is urgent when the probability of "Yes" reaches 0.3771 (the adapter metadata's threshold for English notifications). The system turn is the metadata's instruction followed by "The user's locale is en_US.", and the user turn is *"[Notification]\nTitle: ...\nCommunicationFlag: False\nBody: ...\n\nIs this urgent?"*, read from `SummarizationKit`. Apple's logged token counts match this prompt exactly when the text mentions a time of day and are two tokens longer otherwise, a difference that remains unexplained. On 55 everyday notifications our adapter gives Apple's verdict for 53, the base model for 48 and the alternative decode for 46. On 30 further notifications, selected because these three variants disagree, the adapter agrees with Apple on 28, the base model on 9 and the alternative decode on 7; on the 18 of them whose prompt length matches exactly, the adapter agrees on all 18. The remaining 24 adapters share the file format and the decode, but belong to features that only Apple's own entitled processes can run.

IFP levels. ifp1 and ifp2 use the first 5 and 14 always-on experts plus as many routed ones. In llama.cpp these are views of the shared FFN; at 256-aligned offsets they are whole Q2_K blocks. The level is read from the active adapter (`adapter.afm.ifp`). The level matters: run at ifp2 or ifp3 instead of its own ifp1, the translation adapter agrees with Apple less often.

11. The speculative-decoding draft model

Apple pairs the model with a small draft model for speculative decoding: asset `fm_api_generic.draft`, `model_config v11-ifp-9m`. Its metadata asks for 7 draft tokens per step, exact matching (`draft_soft_match_tolerance` 0) and no adaptive length. The asset has no weights file: everything is inside its Neural Engine program (`binary_0.hwx`, 22 procedures: 3 readouts of 8/16/64 tokens and 19 `extend` variants) and its `program.odix`.

Shape	12 layers, $d = 256$, 8 Q / 4 KV heads $\times$ 32, SwiGLU FFN 768; 9.4 M parameters plus the int4 embeddings
Layer	pre-norm: $h = \mathrm{rms}(x)g_1$; $q, k =$ RoPE (half-split, base $5 \cdot 10^5$) *then* per-head RMSNorm and gain; $x \mathrel{+}= W_o\,\mathrm{attn}$; $x \mathrel{+}= \mathrm{FFN}(\mathrm{rms}(x)g_2)$
Weights	4-bit palettes: each projection is 16 units (one per ANE core), each a sorted 16-entry fp16 codebook, then input-major nibbles; no row scales. Gate tiles carry the 128-byte ANE SiLU table
Vocabulary	100,000 of the 262,144 tokens (a sorted id list in `program.odix`); 95.9% of Apple's answer tokens are in it
Embeddings	int4 codes for the 162,144 other tokens, then the 100,000 draft tokens, in the base model's 8-token interleave, times one per-dimension fp16 scale s; no per-token scale
Readout	tied: the readout procedure's constant tile is $[\,0 \mid s\,]$, i.e. $h \cdot s + 0$, then the int4 matmul over the draft vocabulary

Table 7. The draft model as recovered from its Neural Engine program.

Reading the dataflow. The 132 weight convolutions (11 per layer) are the tasks with Neural Engine opcode `0x46` and a kernel offset. Their roles follow from the surrounding tasks. In layers 1–11 the first 128-channel projection is written straight to the cache by a tile-destination DMA, so it is V. The second passes through RoPE, a 32-wide RMS reduction and a gain multiply before its cache write, so it is K. Layer 0 orders them q, k, v. The attention score scale $1/\sqrt{32}$ sits in the QK matmul's output-scale register, and the softmax is an exp2 with a $\log_2 e$ prescale. The 768-wide FFN hidden is laid out by L2 address: the three up and three gate tiles are written to descending addresses, so the hidden is $[\mathrm{tile}_2 \mid \mathrm{tile}_1 \mid \mathrm{tile}_0]$. In the last layer the compiler re-banks these buffers and leaves one destination register implicit, so we use the conv order there. Norm gains come from `__TEXT,__const`, one 0x4000-byte block per procedure, addressed by the tile-source offsets. In layer 2 the query gain multiply reuses the tile the key gain loaded; the compiler had merged the two. The query and key gains of a layer agree up to sign flips inside rotation pairs and a few ulps, so the norm cannot be moved before RoPE. The final norm computes $\mathrm{rsqrt}(\mathrm{mean}((0.1x)^2) + 1.19 \cdot 10^{-7}) \cdot 0.1$, an fp16 overflow guard equivalent to $\epsilon = 1.19 \cdot 10^{-5}$.

Validation. A NumPy forward of this description predicts Apple's own next answer token (from the 3B model, teacher-forced) 42.95% of the time on the 40 prompts (1127 tokens) and 37.7% on the 230 (6572 tokens). Every alternative tested collapses: cyclic instead of unit-blocked output rows 0.0%, reversed RoPE direction 16.7%, query head $h \to$ KV head $h \bmod 4$ 6.7%, K and V swapped 1.5%, FFN tiles in conv order 2.3%, raw codes without s 0.3%. The input scale is a sharp optimum at 1 (0.25: 27.6%, 4: 36.5%). The ANE SiLU table changes nothing (42.95% either way). Simulating a 7-token chain with exact match, the draft's proposals are accepted 0.61 tokens per step on the 230 prompts, so each target call yields 1.61 tokens.

Checked against Apple's own draft. Apple's draft program does not run from our process, but Apple's runtime reports what it did. After every output token the inference service logs running totals (`log stream`, subsystem `com.apple.tokengenerationinference`, category `SpeculativeDecode`): `countTotalAcceptedDraftTokens`, `countTotalRejectedDraftTokens`, `speculatedComputedCount`. A rise in the first at output token n means Apple's draft had proposed token n; a rise in the second means it had not. That is one bit about Apple's draft per decision, at a known position in a token sequence we also have. Read as a greedy chain the bits did not fit: Apple accepted 53% of the decisions where our draft's top token was right only 40% of the time, and the two agreed on 78%. The counters never speculate deeper than two tokens, the draft's programs take 8 tokens, the metadata sets `draft_steps` to 7 and `draft_use_maximum_likelihood_tree`,

and when Apple accepted, the token was among our draft's five best in 97.5% of the cases with a sharp edge after the fifth.

Apple therefore speculates with a *tree*: each round scores the last real token and 7 draft nodes, the 7 most likely nodes of a depth-2 tree by joint probability, and a token is accepted if it is a node at its level. Replaying that tree with our draft on Apple's answers (60 prompts, 1435 logged decisions, 1331 of them more than one token away from a program switch, see below) gives the same decision as Apple's draft in 95.6% of the cases, with the same acceptance rate (55.7% against 55.9%); a tree of 6 or 8 nodes breaks the rate (53.6%, 57.3%), and an unrelated model with that rate would agree 51% of the time. The remaining disagreements are near-ties: where the token's joint probability is within a factor $2^{1/4}$ of the tree's threshold the two agree 60% of the time, beyond a factor 2 they agree on 99.2% (1134 decisions), beyond a factor 4 on 99.7% (951). The GGUF run by llama.cpp gives the same greedy decisions as the NumPy forward on all 1435.

Two details of Apple's runtime are visible as well. When the sequence reaches 96 or 128 tokens the runtime moves to the next larger fixed-context program, and that round is logged as rejected whatever the draft proposed (36 such rejections of tokens our tree contains); the 104 decisions within one token of those lengths are therefore left out. And the gain of Apple's scheme is larger than a chain's: 0.56 of the tree's decisions are accepted.

In llama.cpp. `afm2gguf.py --draft` writes the draft as architecture `afm-draft` (61 MiB): the codes as Q4_0 with $d = 1$ (exact), s as `token_embd_scale` and `output_scale`, F16 projections (palette values are fp16), and an `output.bias` of $-\infty$ outside the draft vocabulary. The bias matters: the best draft logit is negative at about 1% of positions, so zero rows would sometimes win. On Metal, its logits match the NumPy reference to 0.03.

Speculative decoding places a requirement on the target graph. A verify batch [last token, 7 draft tokens] is several tokens of one sequence, which would otherwise be treated as a prompt chunk: one shared expert pick, persisted, and no input bias. Verify batches are therefore recognised (every token requests logits and the run does not start at position 0) and processed exactly as single decode tokens: input bias except at re-pick positions, and routing on the latest re-pick token at or before each token. Draft tokens can be rejected after they have re-picked, so the routing state keeps two picks per sequence, each tagged with its source position. A pick from a position the batch starts at or after belongs to rolled-back tokens and is ignored. With this handling, 8-token verify batches reproduce one-token-at-a-time decoding to $2 \cdot 10^{-5}$ in the logits.

Lossless, but not faster in llama.cpp. With greedy sampling, 7 draft tokens and no confidence cutoff (Apple's settings), `llama-server` with the draft returns exactly the plain answers on all 40 prompts, on Metal and on the CPU, and the plain answers are unchanged by the patch. About 11% of drafted tokens are accepted, as the simulation predicts. Speculation is nevertheless slower in llama.cpp. Measured back to back, it was 2.1× slower on Metal and 3.5× slower on the CPU (1.8× with 3 draft tokens). A verify batch costs several single decodes here, because the 2-bit expert matmuls are compute-bound and every draft step has a fixed per-call cost. Apple's Neural Engine runs fixed-shape 8-token `extend` programs, so a verify step costs about as much as one token, and 0.6 extra tokens per step is a real gain there. `tools/eval/speculative.py` reproduces these measurements.

12. The image encoder

Images reach the model through a separate asset, `instruct_3b.image_encoder` (`model_cls_name ImageTokenizer`). It holds one Neural Engine program with six procedures, one per patch count $N \in \{1152, 1728, 2304, 3600, 4608, 5760\}$, and `swift_preprocess_weights.bin`. The program's inputs are the patches $[N, 3, 14, 14]$, an additive position embedding $[N, 1024]$, RoPE rotation matrices $[N, 32, 2, 2]$ and an additive key mask. Its output is $N/9$ embeddings of

width 1536, which the 3B model reads between `<start_of_image>` and `<end_of_image>`. Apple's runtime log names the sizing rule: the aspect ratio is kept, the image is scaled to at most 338 688 pixels (1728 patches), and both sides are snapped to multiples of the 12-patch attention window.

Shape	ViT, 24 layers, $d = 1024$, 16 heads $\times$ 64, SwiGLU 2816; 358 M parameters with the merger
Block	$h = \mathrm{rms}(x)g_1$; q, k, v with bias; RoPE then per-head RMSNorm and gain on q, k; attention inside 12×12-patch windows, full attention in layers 7, 15, 23; $x \mathrel{+}= W_o a + b_o$; $x \mathrel{+}= \mathrm{SwiGLU}(\mathrm{rms}(x)g_2)$
Positions	learned 16 × 16 table (bicubic-resized to the patch grid) added after the patch embedding, then a pre-norm; 2D RoPE on interleaved pairs: pair $2f$ rotates by the row, pair $2f{+}1$ by the column, both at $\theta_f = 10000^{-f/16}$
Token order	window-major; inside a window the 3×3 merge groups in raster order; inside a group the patches in raster order
Merger	final RMSNorm; 9 consecutive tokens concatenated (9216) $\to$ 4096 $\to$ LayerNorm ($\epsilon = 10^{-6}$) $\to$ GELU $\to$ 2048 $\to$ 1536, no biases
Weights	int4 (16-entry LUT × fp16 row scale) except the patch embedding and the last projection (fp16)

Table 8. The image encoder as recovered from its Neural Engine program.

Reading the weights. The int4 blocks resemble the 3B model's, but the header length depends on the convolution's flags: a 16-entry fp16 LUT, 32 zero bytes, the row scales, and the row biases *only if* the convolution has a bias (Neural Engine register 1, bit 0x10). A 128-byte activation table precedes all of this when the convolution applies an activation (bit 0x20000). A fixed 128-byte header is correct only for 16-row blocks with a bias; under that assumption every bias-free block (all FFN and merger weights) is read 4 to 12 inputs out of place, with its last inputs taken from padding. The block sizes look the same either way because each unit is padded to 64 bytes. Compiling known weights with Apple's compiler settles the question: a 9216→128 projection, a bias-free up projection and a SiLU-gated projection decode exactly with the flag rule and at chance level (6%) with the fixed header.

The effective bias is the stored bias times the row scale, and times 2^{e-32} when the convolution's task writes the Neural Engine bias register `NEBias` with exponent field e (its value field is 0). In this encoder only the V convolutions of layers 18 and 19 do so ($e = 31$); missing it left their biases twice too large, which costs 0.001–0.007 of agreement with Apple's encoder. Projection tiles are written to descending addresses, so channel order is the reversed tile order. The weights are split over two kernel sections; each block is assigned to the section where its LUT is valid. That rule is unambiguous for 1293 of 1297 blocks, and the remaining 4 follow their neighbours.

Norm gains are [bias | scale] constant tiles that the planar engine fetches; we take every gain from its fetch offset rather than from a layout formula, because the global-attention layers use a different constant layout. The compiler also folded a 0.01 pre-norm scale into the encoder: the last residual add scales its residual operand by 0.01 (a planar-engine operand scale, the same register that holds 1/8 in the global layers), and 0.1 is folded into each of the last layer's up and down projections, which are ten times smaller than every other layer's. The converter undoes this. Activations are 33-knot piecewise-linear tables: SiLU on $[-8, 8]$, GELU on $[-4, 4]$, with explicit linear tails.

Ground truth from Apple's own encoder. Unlike the language model, the encoder's MPSGraph executable runs on the Neural Engine from our own process (about 0.8 s per call), so Apple's encoder is a direct oracle for any input we build. A copied package does not load ("Program load failed"), so weights cannot be patched to bisect. We therefore used constructed inputs. When all N tokens are identical, every attention layer returns V, so the encoder becomes a per-token function of one

1024-vector; and the merger can be inverted from the output. With the fixed header the inversion stalls at cosine 0.991 for every seed; with the flag rule it reaches 0.99999. The NumPy forward then matches Apple's output at cosine 0.987 on constant-token inputs and 0.95–0.97 per token on real images. One further correction is invisible to constant-token inputs: layer 0 emits its key projection before its query projection. With q and k of layer 0 exchanged, the match on real images rises to 0.98–0.99 per token. The llama.cpp projector requires two further measures to follow: Apple's activation tables in place of the exact SiLU and GELU (with exact activations it matches Apple's encoder at 0.975), and an antialiased bicubic resize, since a plain 4-tap bicubic aliases when a large photo is reduced (0.84 against Apple's encoder for a 4096×2160 photo, 0.985 with antialiasing). From the image files the projector matches Apple's encoder at 0.98–0.99, its Metal and CPU paths agree to 0.9999, and 6 of 13 free-running captions are identical to Apple's.

Every wiring choice that can be varied has a sharp optimum: the alternative V/O and gate/up/down pairings score 0.16–0.23, and in every layer each other choice of V group is worse; on a real image, swapped q/k scores 0.09, half-split RoPE pairs 0.50, x-before-y 0.51 and transposed patches 0.85. Fitting per-layer branch scales gains less than 0.4 points, so the remaining percent is not a wiring error. A Core ML replica of our forward run on the Neural Engine was less faithful still (0.76), so we attribute the gap to the numerics of Apple's compiled program.

Through the 3B model: the RoPE layout. Our prompt places the question, a newline and the image in the user turn, 266 tokens in total, exactly the count Apple's runtime logs for the same request. Several things the encoder program cannot tell us are chosen by Apple's host code: the pixel normalisation, the position table's interpolation, and how the 32 RoPE rotations of a head are split between the row and the column. Comparing against Apple's encoder cannot settle these, because both encoders receive whatever inputs we build. The 3B model can. With the obvious layout (row on pairs 0–15, column on 16–31) the language model received colours and scene gist but no geometry. Shuffling the 192 image tokens hardly changed anything, and asked whether a red circle is left or right, top or bottom, it answered "Left" and "Top" for all four pictures, where Apple's model answers all four correctly.

The correct layout *alternates*: pair $2f$ rotates by the row and pair $2f{+}1$ by the column, both at frequency θ_f. With it all four spatial answers are right with margins of 5–12 logits. The mean log-probability of the key words of Apple's captions (circle, HELLO, equation, Golden, balloons, ...) rises from -2.99 to -0.20. The other host choices are then sharp: a negated rotation mirrors the image ("HELLO" at -8.2), bilinear instead of bicubic position interpolation is worse (-0.26), and so is a transposed position table (-0.21). ImageNet normalisation is slightly better than CLIP's (-0.20 vs -0.22). With CLIP's statistics a plain white image is barely "white" (0.2-logit margin over gray, against 6.5 with ImageNet's), and CLIP's mean vector does not occur anywhere in the operating system's shared cache.

Teacher-forced on Apple's captions of 13 images (168 tokens), our encoder gives the 3B GGUF Apple's next token 94.0% of the time (NLL 0.29), Apple's own encoder on the same inputs 92.9% (NLL 0.29), and no image 58.9% (NLL 2.11); these figures were measured before the layer-0 correction above. Free-running through llama.cpp with Apple's instructions, 6 of the 13 captions are identical to Apple's character for character, among them "The word "HELLO" is written in black capital letters on a white background.", and the rest are close paraphrases ("The image shows a math problem: "2 + 3 = ?"." for "...a mathematical equation: 2 + 3 = ?.").

In llama.cpp. `afm2gguf.py --vision` writes the encoder and merger as a multimodal projector (685 MiB, F16), projector type `afm` in llama.cpp's `mtmd` library. Tokens are processed in Apple's order, with a window mask on 21 of the 24 layers. The query and key head dimensions are permuted so that llama.cpp's vision RoPE (rotating dimensions j and $j{+}32$, row for $j < 16$, column above) reproduces Apple's alternating interleaved pairs. Its embeddings match our NumPy forward at cosine 0.990. On an M4 an image encodes in 0.66 s on Metal.

13. The 300M base model

Apple ships a second, smaller language model, display version `afmplus-v11.0-pico` (asset `com.apple.fm.language.instruct_300m.base.generic`). Fifteen other assets on the same Mac carry the `instruct_300m` name, among them the safety guard (`instruct_300m.safety`), `misc_safety`, `mm_guard`, `action_validator` and several classifiers. In a memory capture of Apple's runtime, the 300M model's input buffer also held a guard turn ending in the model answer `'safe'`. The asset holds one Neural Engine program with 22 procedures (11 with rank-32 adapter slots, 11 with rank-64) and `program.odix` with the embedding.

Shape	24 dense layers, $d = 1024$, 16 query / 4 KV heads × 64 (query head h reads KV head $\lfloor h/4 \rfloor$), SwiGLU 3200; 299 M parameters plus a tied 262 144×1024 embedding
Block	the draft model's layer (Section 11): $h = \mathrm{rms}(x)g_1$; RoPE then per-head RMSNorm and gain on q, k; attention scale 1/8; $x \mathrel{+}= W_o a$; $x \mathrel{+}= \mathrm{SwiGLU}(\mathrm{rms}(x)g_2)$
Positions	RoPE base 500 000, half-split; the cos/sin table ships in `program.odix` directly after the embedding
Embedding	int4 codes × one fp16 scale per dimension; readout $\mathrm{rms}(x)\, g_{\mathrm{out}}\, s$ against the same codes, plus a mask of −40000 on 144 unused ids
Weights	int4: 16-entry fp16 LUT × fp16 row scale; 960 blocks (40 per layer)

Table 9. The 300M base model as recovered from its Neural Engine program.

Decoding. The model is read as the draft model and the image encoder are: from the Neural Engine task stream rather than from symbol names. Five properties of the format are essential, several of them shared with the image encoder; a decode that misses them recovers every tensor yet produces no text. (1) The int4 header is 96 bytes (LUT, 32 zero bytes, 16 row scales; the bias flag is clear), followed by the codes and 32 bytes of zero padding; an assumed 128-byte header shifts every input axis by 4 positions (12 for the 128-row blocks). (2) The norm gains are not folded away: the planar engine fetches all 97 of them (four per layer and a final one) from `__TEXT,__const`, beside a $[0 \mid s]$ tile that scales the hidden state before the readout and the −40000 vocabulary mask. (3) The embedding is a plain 8-token interleave starting at the data section; an apparent lane "skew" and 4-row shift are artefacts of a wrong base address. (4) The per-layer multiplier (9.5–12.0) applied with the key gain is not a temperature. It is the int8 scale of the key cache, and the $q{\cdot}k$ matmul's own fp16 scale register divides it back out: their product is 0.1250 in all 24 layers. Values are likewise stored in int8 (scales 305 down to 10.5) and rescaled in the attention-times-V matmul. (5) The two 256-wide projections are emitted in different orders: layer 0 computes K first, every later layer V first, as in the draft model. We assign them by dataflow: K is the one whose output passes through RoPE and the head norm. Block order within a tensor follows the destination addresses (the query blocks and the up projection's are written to descending addresses), as for the encoder.

Results. Apple exposes this model through no public API, so its outputs cannot be compared as for the 3B model. We evaluate it as a language model instead. On two passages of plain English (172 and 102 tokens) its perplexity is 38.2 and 24.6, the median rank of the correct next token is 2, and it ranks it first 29% and 39% of the time. On random token sequences repeated twice it copies the second half 78% of the time, the signature of trained induction heads. Each of these properties is necessary: with a positional K/V assignment induction is 0% and perplexity above 100 000; swapping K and V in layer 0 as well leaves perplexity above 20 000; halving or doubling the logits raises it threefold to ninefold, so the readout is calibrated as decoded. Greedy decoding through llama.cpp:

> The capital of France is → *Paris.* Water boils at → *100°C.*
> Once upon a time, there was a little girl named → *Lily. Lily was a little girl who loved to*

play with her toys…
(chat turn) What is 12 plus 7? → *12 plus 7 is 19.* Write one sentence about the ocean. → *The ocean is a vast, mysterious expanse of water that stretches for miles, teeming with life, diverse ecosystems, and endless horizons.*

In llama.cpp. The layer is the draft model's, so `afm2gguf.py --pico` writes it with the `afm-draft` architecture and no new runtime code: the embedding as Q4_0 with unit scales (exact), the projections as F16, the full 262 144-token vocabulary with Apple's mask as the output bias (721 MiB). Its greedy continuations are identical to our NumPy forward's.

Adapters and the safety guard. Every 300M adapter, including the default one inside the base package (`lora_32/64_constant_data.bin`), is a flat file of fp16 (A, B) pairs, A as $[r, \text{in}]$ and B as $[\text{out}, r]$, per layer in the order attention output, q, k, v, `linear_0`, `linear_1`, FFN output. The layer order requires care. The pairs follow the package manifest's tensor names sorted as strings, so the layers come as 0, 10–19, 1, 20–23, 2–9. Read in numeric order, every adapter looks inert (the guard below never says "safe"); with a plain numeric-string sort (0, 1, 10, …) the adapters work only partially. `linear_0` is the gate (log-probability of the right first token -0.10 against -0.55 as up), the Neural Engine program scales the product by 0.5, and every role contributes.

The default adapter. With the correct order, the base package's own rank-64 adapter is a trained default: it lowers the base model's perplexity from 38.2 to 30.8 and from 24.6 to 21.4 (llama.cpp, 128-token windows: 26.9 to 22.1). Twelve task adapters of rank 32 or 64 ship on the test machine, and all share its starting point: their weights correlate 0.33–0.99 with the default adapter of the same rank, where unrelated adapters would correlate near 0.

The safety guard. The text safety guard (`instruct_300m.safety`, rank 64) carries Apple's instruction: answer `safe`, or one line per problem of the form `category,mention` or `category,action` over 14 categories (hate_harassment, violence_gore, self_harm, jailbreak, vulgarity, …). Run through llama.cpp with that instruction as the system turn, it answers `safe` for every benign prompt and answer tested, and labels harmful text in the requested format: `self_harm,action` for a plan to self-harm, `jailbreak,action` for "Ignore all previous instructions…", `vulgarity,action` for swearing. It separates talking about a thing from doing it: a history of the battle of Verdun gets `violence_gore,mention`, a friend's past suicidal thoughts `self_harm,mention`, a news report of an arrest `safe`. The smaller `mm_guard` (rank 32, nine categories) agrees on everything except the swearing, which it calls `safe`; vulgarity is not one of its categories. `afm2gguf.py --pico-adapter NAME` writes any of them as a llama.cpp LoRA with Apple's instructions in its metadata, and `tools/afm300m_classify.py` runs a classifier as Apple renders it.

Checked against Apple's guardrail. Apple's public FoundationModels API runs the same guard: it rejects a request with "May contain unsafe content" when the guard's labels map to `unsafe` under the API's row of Apple's policy table (`feature_output_category_mapping.json`: eight labels such as `violence_gore,action` or `self_harm,action` block; mentions, jailbreak attempts and vulgarity do not). On 26 prompts (benign questions, mentions, nine harmful requests, two jailbreaks, two vulgar ones) our guard with that table predicts Apple's decision on all 26, including the cases our guard labels but Apple's policy lets through. On a second set of 74 prompts (32 of which Apple stops) it predicts 69; the five misses are borderline (four insults or indirect threats that ours labels as an action and Apple lets through, and a hobby question about lock picking that Apple stops), so 95 of 100 in all. One request Apple stopped at its output check ("Response may contain sensitive…"), which our guard already flags on the input.

The image encoder. The 300M model has its own image encoder, `instruct_300m.image_tokenizer` (`ImageTokenizer`, model id `afm-plus-v11-300m-instruct-4bit-pal`).

Its program has the 3B encoder's structure exactly: 24 layers, windows of 12×12 patches with full attention in layers 7, 15 and 23 (counted from the attention matmuls per layer), the same 3×3 merger, and a last projection to 1024 instead of 1536. Its position table differs, so it was trained separately.

Three details differ from the 3B encoder. The weight palettes are sorted real-valued codebooks rather than integers; there is no folded 0.01 scale (the last layer's weights are in line with the others); and 15 of its 96 attention bias tensors come from tasks that write the `NEBias` exponent ($e = 28$–31), so their biases are 2^{e-32} times the stored value. Read without that factor they are 2–16 times too large, which no weight statistic reveals but Apple's own encoder does: on identical inputs the agreement is cosine 0.45–0.54 per token without the factor, and 0.61–0.78 on photos and 0.97 on constant inputs with it. Two further corrections, both checked against Apple's encoder, bring it to 0.68–0.87: layer 0 emits its key projection before its query projection (as in the 3B encoder), and the five final-norm gains that are fp16 subnormals are taken as zero.

The merger normalisation. The remaining discrepancy is a single layout property of the merger. Its normalisation is a depthwise convolution whose kernel holds a (bias, gain) pair per channel in 16 units of 256 pairs, and the units are *interleaved* over the channels: channel c is entry $\lfloor c/16 \rfloor$ of unit $c \bmod 16$. Reading the units one after another puts every gain and bias on the wrong channel. Nothing in the bytes indicates this, and the 3B encoder is nearly insensitive to it (0.992 to 0.994 on constant inputs), because its gains and biases are nearly uniform; the 300M encoder's are not. It was identified with constant-token inputs, where Apple's encoder is a per-token function: the reconstruction's output was uniformly 17% too small in norm, the 3B encoder's was not, and refitting only the 4096 biases of that layer on 54 inputs raised held-out agreement from 0.956 to 0.997, and on photos (never fitted) from 0.68–0.87 to 0.95–0.97. The fitted biases correlated with the stored ones only under the transposed unit order. With that order and no fitted quantity the 300M encoder agrees with Apple's at 0.998 on constant inputs and 0.990–0.999 per token on photos (the 3B encoder: 0.989–0.997), and llama.cpp's projector at 0.98–0.99 from the image files.

The two programs were also compared register by register: they are identical task for task except for the bias exponents, the folded 0.01, and the output width, and every weight block and bias tensor of one correlates with its counterpart in the other ($r \approx 0.98$ for weights, slope 1.00 for all 96 bias tensors under the exponent rule). The same rule explains every one of the 96 tensors' power-of-two ratio to the 3B encoder's biases. The same decoder therefore covers both (`afm2gguf.py --pico-vision`); the 3B projector it writes is unaffected.

Image classifiers. In llama.cpp the encoder runs as the existing `afm` projector. The only runtime requirement is that the `afm-draft` graph apply the per-dimension embedding scale to text tokens only, so that image embeddings enter unscaled. With the default adapter the 300M model describes images ("The image contains the text "HELLO" in a stylized font."; "a blue square with a white background"). The image classifiers are adapters of the same kind as the guard: `misc_safety` holds Apple's instructions for violence/gore classification and for counting people ("0, 1, 2"), and `prepubescent_safety` checks for children shown with adult features. On stock photos from Keynote's themes the people counter answers 0 for all four photos without people, 1 for all five single portraits (without the bias rule it misses a person whose face is hidden by a hood), 2 for two women, and 2 for three boys, where 2 is the largest answer allowed. The two safety classifiers answer `safe` for every benign image tested, including synthetic red paint splatter; they were not tested on harmful images.

The following were excluded as causes of the residual difference between the two encoders: the Neural Engine's softmax (its 33-knot 2^x and $2/m$ tables and fp16 $\log_2 e$), fp16 saturation in the norms, reduced-precision palettes, and the kernel-section choice of the three blocks that are valid in both sections; none changes the agreement. Apple's encoder is also numerically stable: perturbing an input by 10^{-3} moves its output by less than 0.001 in cosine.

14. Fidelity

The reference is Apple's own model, queried through the public FoundationModels API at temperature 0 with the instructions "You are a helpful, concise assistant." (`tools/afm.swift`) on 40 prompts. We measure two things: *teacher-forced* top-1 agreement (Apple's answer is fed in and we count how often our most likely next token is Apple's; 1127 answer tokens) and *free-running* agreement (our own greedy answer compared with Apple's, character for character).

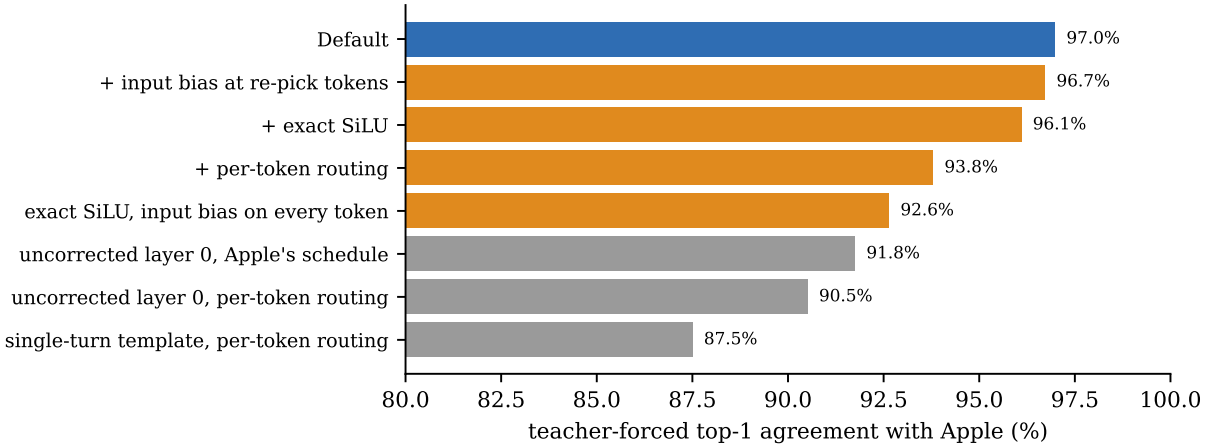


Figure 11. Teacher-forced top-1 agreement with Apple's greedy answers (40 prompts, 1127 tokens). Blue: the default configuration (Apple's schedule, the Neural Engine SiLU table, no input bias at re-pick tokens); the three rows below it each change one further property. "Uncorrected layer 0" denotes an embedding scaled by 100 and the input bias on every token. The exact template is worth three points, the layer-0 corrections (no embedding multiplier, input bias on generated tokens only) four more, and Apple's routing schedule outperforms per-token routing.

Configuration (exact template)	teacher-forced	first token	identical answers
Default	**96.98%**	38/40	17/40
Input bias also at re-pick tokens	96.72%	38/40	17/40
Exact SiLU instead of the Neural Engine table	96.10%	37/40	17/40
Exact SiLU, per-token routing	93.79%	36/40	12/40
Uncorrected layer 0 (embedding ×100, bias on every token)	91.75%	35/40	7/40
Uncorrected layer 0, per-token routing	90.51%	38/40	11/40

Table 10. Configurations. The default uses Apple's schedule, the Neural Engine SiLU table and no input bias at re-pick tokens; each row above the rule changes one property relative to the row before it. "Identical answers" = our free-running greedy answer equals Apple's exactly (up to 200 tokens); one different early token sends the rest of an answer elsewhere, so this measure is dominated by the first few tokens.

Ablation (10-prompt set, single-turn template)	top-1
always-on + 23 routed experts	85.6%
always-on experts only	62.9%
routed experts only	1.4%
16 instead of 23 routed / 14 routed	83.5% / 82.1%
standard (pre-norm) residual instead of x/y	0%

Table 11. Ablations: both expert groups are needed, the routed count matters, and the residual form is essential.

The layer-0 input: no multiplier, and a bias only for generated tokens. An initial analysis fitted an input multiplier k and found the score flat for $k = 8$–14. That scan was invalid: llama.cpp's generic embedding code already applied the GGUF's `embedding_scale`, and the AFM graph applied it again, so the scan really covered $k = 80$–140. Dumping llama.cpp's intermediate tensors and comparing them with an independent NumPy forward revealed a factor of exactly 10 at the layer-0 input. Scanning the true multiplier with the expert picks held fixed gives a sharp optimum at $k = 1$ (mean log-loss of Apple's tokens 0.140, against 0.143 at 0.75 and 1.25 and 0.240 at 100). This agrees with Apple's program: the embedding gather has no multiply.

With $k = 1$ the body of the network and the router disagreed. The body wanted the input bias added, but the router's picks were best when the bias was negligible. This was established by computing the picks from a forward at a larger k, or with the bias removed; a version that changed only the layer-0 attention balance did not help (78%). Apple's ANE program has two procedure families. The prompt procedures (`prompt_opt`) add a zero constant at layer 0. The extend procedures, which compute generated tokens, add the bias vector. Applying the bias to generated tokens only is one consistent model for both parts. It reaches 95.9% with the picks held fixed and 96.1% in the llama.cpp runtime, which gives the bias to single-token decode steps and not to prompt chunks. Flipping the bias's sign costs two points, so the vector is genuine; it belongs to one procedure family only.

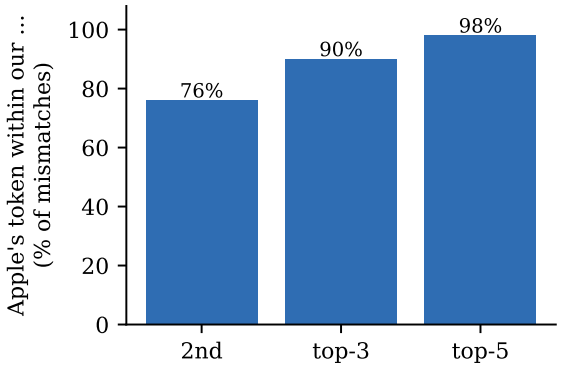


Figure 12. Rank of Apple's token at the 93 disagreements of the uncorrected-layer-0 configuration: Apple's token is our second choice in 76%, in our top three in 90% and in our top five in 98%; the median logit gap to it is 1.15, and 45% of the misses are within one logit.

The tokenizer. Apple's `SystemLanguageModel.tokenCount` gives Apple's own token count. A tokenizer with byte-fallback tokens labelled as normal text and control tokens mislabelled matches it on only 73 of 81 texts. The converter's tokenizer matches on 81/81 (and on 16/16 texts built from tabs, runs of spaces and indented code) and raises fidelity from 86.0% to 87.5%. Apple's tokenizer

file marks all user-defined pieces with one type code; mapped naively, llama.cpp prints space runs as the escape character U+2581 instead of spaces and drops tab runs entirely, so the converter types space runs as normal text and tab runs as user-defined.

15. The converter

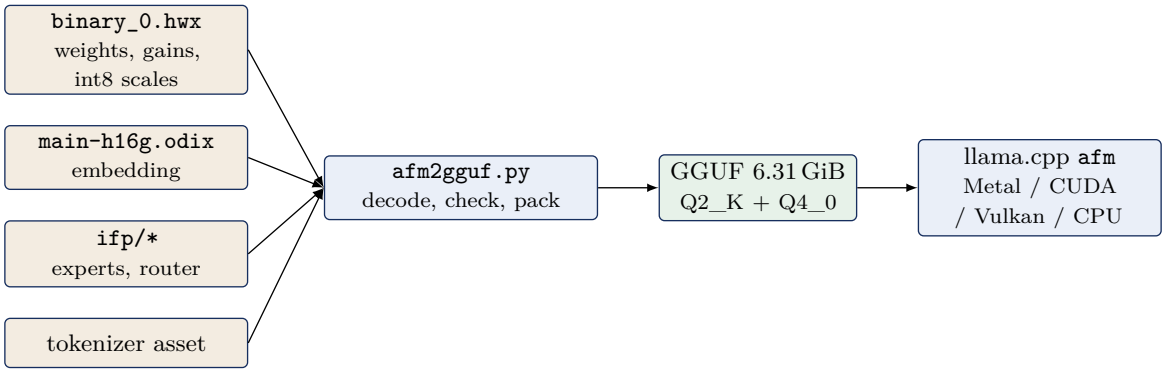


Figure 13. Conversion runs on the Mac that holds the assets (2–3 minutes); the GGUF then runs anywhere.

`afm2gguf.py` needs only NumPy and gguf-py and contains no Apple data. It finds assets by their `AssetSpecifier`, decodes everything directly, and refuses to write if a check fails: palette and container layout, 3504 units in 219 chunk-sets, the router mask, the tokenizer layout, the embedding scale array (chosen by row-norm consistency) and—decisively—the gain-table alignment, chosen by the FFN coupling-sign agreement (0.99 for the true start, 0.59 for the one-vector-early candidate that passes every shape test). Its output is byte-identical, tensor by tensor, to the export used for the measurements in this report. With `--draft`, `--vision`, `--pico`, `--pico-vision` and `--pico-adapter` it writes the draft model, the image encoder, the 300M base model, its image encoder and its adapters instead; for the image encoder, every gain is taken from the program's own constant fetches; the 300M model's RoPE base is fitted from the table in its own asset (500 000).

16. Negative results

The following approaches were unsuccessful. They are recorded to spare others the effort.

- Running Apple's backbone on the ANE from our own process: the kernel refuses LLM base-model requests from non-platform binaries, even with AMFI disabled.
- More than ten forward variants tried before the gain alignment was corrected: Gemma-style $1+g$ gains, standard residual, swapped or negated RoPE tables, embedding scales, alternative row stripes, GQA maps, within-vector gain permutations, per-row scale permutations—all 0% top-1, because the defect was elsewhere.
- A LoRA adapter as the missing component: the Foundation Models API's generic adapter for v11.2 is untrained (B RMS $3 \cdot 10^{-5}$), so the base model is already what the public API runs. Its "signature" (the SHA-256 of the empty string) proves nothing, because the trained feature adapters carry the same one (Section 10).
- Apple's `MODLM current context` N `asked to run` K `tokens` log as an oracle for draft acceptance: it interleaves the guard model, the target and the draft (and several generations) with no stream id, and the draft's catch-up calls are only ever 1 or 3 tokens, so per-step acceptance cannot be read off it. (The `SpeculativeDecode` counters can; see Section 11.)
- Patching Apple's image encoder to bisect it: a copied MPSGraph package does not load on the Neural Engine. A Core ML replica of our encoder on the Neural Engine was less faithful (0.76) than NumPy (0.987).

- Before the alternating RoPE layout was identified, we attempted to restore spatial understanding with RoPE position rescaling (the RoPE table has 481 rows; $\times 10$ or fit-to-480 positions), window-relative positions, other window, group, patch and output orders, cropped position tables, collapsed LLM positions for the image tokens, bidirectional attention over them, rescaled embeddings, no int8 K/V emulation and image-before-question prompts. None helped; each leaves the axis layout unchanged and therefore could not.
- Weight statistics such as "alignment ratios" and residual-collapse measures: uncalibrated, and on a healthy model indistinguishable from AFM.
- Decoding the 300M model from symbol names, compiled probes and permutation searches: these recover the tile element order exactly but never produce text. The conclusions they suggest—no norm gains ship, the embedding has a lane skew and a 4-row shift, the residual-writing projections hide an unreadable layout, some layers' FFNs "hurt"—are all consequences of the assumed 128-byte header, the wrong embedding base and the positional K/V assignment (Section 13).
- Reading the 300M adapters' layers in numeric order. Every adapter then appears inert, and the base package's default adapter appears slightly harmful, as a placeholder would. The files follow the manifest's string-sorted tensor names (Section 13).
- Pooled expert routing rules (mean of inputs, of logits, of probabilities; prompt or user turn; prefix mean, vote counts): all below per-token routing, and all wrong—Apple routes from a single token (Section 8). They appear plausible only under an incorrect prompt template.
- Routing from the layer-12 FFN input instead of its attention input: 89.4% teacher-forced, and three times too many experts changed per re-pick (109 versus Apple's 36).
- Reading Apple's between-prompt expert changes from the cache log: those counts (361–556 of 1012) include cache eviction while the service is idle and do not measure pick differences; only re-picks within one generation do.

17. The remaining gap

The prompt template, the expert schedule, the layer-0 input and the activation function are Apple's own. 37 of 1127 answer tokens still differ, and Apple's token is almost always our second or third choice. Attributing a residual gap to numerical noise requires care: before the layer-0 corrections a gap of eight points could be, and was, mistaken for a noise floor. The following were therefore measured directly:

(1) **The ANE's matrix multiplies are accurate.** We ran our own CoreML programs on the Neural Engine, confirming the device with Core ML's compute plan. Sums accumulate at high precision and only the fp16 output is rounded (relative error 2.8×10^{-4}, the same as fp32 accumulation with an fp16 result). The low-precision accumulation one first sees comes from Core ML's fp16 CPU path, not the ANE.

(2) **Its SiLU is a 33-knot piecewise-linear table** (knots every 0.5 on $[-8, 8]$, fitted values, for example 0.2964 at $x = 0.5$ where SiLU is 0.3112). The same 128-byte table appears 2113 times in AFM's ANE program, once per FFN task, and it is the program's only activation table. We tabulated it for all 65 536 fp16 inputs. It is exactly linear interpolation between the knots, with a left tail clamped to zero below -10.2 and $x - 0.0039$ above 8, rounded to fp16. With the uncorrected layer-0 input, emulating it is neutral (10 fixed, 8 broken) because the layer-0 error dominates. On the corrected model it fixes 14 and breaks 5 tokens (96.10% $\rightarrow$ 96.72%), so it is the default. Its softmax, exp and rsqrt are also approximate (0.3–0.4% relative error).

(3) **Apple's speculative decoding is exact.** Every FM-API adapter sets `draft_soft_match_tolerance` to 0 and turns rejection sampling off, so the draft

model never changes a greedy token. The FM-API adapter is the declared-empty adapter (signature = SHA-256 of the empty string), and there is no repetition penalty.

(4) **The router is robust to arithmetic.** Computing it with fp16 accumulation moves 0.01 of 23 experts per layer-pick, and random noise strong enough to swap a third of an expert per layer changes 2% of predictions without moving the agreement.

Every weight, gain, bias, the router, the tokenizer, the template and the routing rule come from Apple's files and code; nothing was trained or fine-tuned.

Path noise. On the corrected model, the CPU and Metal backends (identical mathematics) disagree on 0.71% of answer tokens (8/1127), and their agreement with Apple is identical (96.72%). The remaining 37 misses are near-ties: Apple's token is our second choice in 84% and always in our top three, and the median logit gap is 0.40. Rounding every matmul output to fp16, as the ANE stores it, gives 96.89% (within noise). With the uncorrected layer-0 input, mathematically identical variants disagree on 0.1–2.4% of next-token choices: fp16-rounded activations against fp32, llama.cpp's RoPE kernel against an elementwise one, or one batch against a prompt batch followed by single steps. The 2-bit weights and the int8 KV cache amplify rounding, so a percent or so of residual disagreement is expected even from an exact model.

A larger test. The 40-prompt set moves by about ±0.4% from noise alone, so we added 190 prompts (6572 answer tokens in all) and re-ran every candidate (Table 12). The Neural Engine SiLU is worth a full point. Every other ANE detail we could emulate is neutral: fp16 matmul outputs, int8 KV on or off, keys quantised before their gain, and fp16 RoPE tables. Truncating instead of rounding the int8 cache costs a point, so Apple rounds to nearest. Of the remaining 316 misses, 2.9% of tokens are missed under any routing and 1.9% only under Apple's schedule. The latter is consistent with our prompt-level pick differing from Apple's by a boundary expert or two: the 23rd and 24th router logits are a median 0.003 apart, about the size of a rounding difference in the layer-12 state. Apple's log also confirms the re-pick timing. Around a boundary the runtime steps one token at a time, and the pick at position 96 uses token 96 itself. Apple's fp16 logits never tie at the top, and its softmax errors (about 0.2% in the denominator) cancel in the post-attention RMS norm.

Variant (230 prompts, 6572 answer tokens)	top-1	fixed / broken
Default (Apple's schedule, ANE SiLU, int8 KV, no input bias at re-pick tokens)	**95.27%**	12 / 7
Input bias also at re-pick tokens (reference row)	95.19%	–
exact SiLU	94.17%	72 / 139
per-token routing	93.34%	–
fp16 rounding of every matmul and attention output	95.13%	30 / 34
int8 KV emulation off	95.07%	46 / 54
int8 rounding by truncation	94.05%	64 / 139
keys quantised before their gain	95.09%	43 / 50
RoPE cos/sin rounded to fp16	94.99%	21 / 34
re-picks that favour already-loaded experts ($\beta = 5 \cdot 10^{-4}$)	95.15%	–

Table 12. Candidates on the larger set, each compared with the reference row (fixed / broken = tokens that become right / wrong).

Running the dense layers on the Neural Engine. As a direct test of the arithmetic, we rebuilt layers 0–11 as a Core ML program, which placed every operation on the ANE: int8 KV round and clip, the ANE's own SiLU and softmax. We fed its layer-12 state to llama.cpp for routing and the remaining 44 layers. The state differs from our emulation by about 15% (relative norm) after 12 layers; in isolation each layer's FFN agrees with our SiLU-table model to fp16 rounding (0.5%), and

int8 steps in the very coarse layer-0 values amplify the differences. Agreement with Apple stays the same: 94.92% against 95.19%, but 224 of 6572 tokens flip (103 fixed, 121 broken). The control, the same program in fp32 on the CPU with exact SiLU, gives 94.42%. So two plausible versions of the hardware arithmetic disagree on about 3.4% of tokens without either being closer to Apple, which is the size of the remaining gap. Core ML compiles our graph differently from Apple's hand-built ANE program, so even the real hardware does not reproduce Apple's rounding. Closing the remaining gap would require that program's exact arithmetic.

Apple's own expert decisions. Apple's runtime logs every pick per layer ("layer N experts: ..."). The expert list is marked *sensitive*, so it stays redacted even with a logging profile that enables private data for the inference service. That profile does reveal the subsystem's ordinary private values. The cache statistics are public, though. At each pick the runtime logs `reordered`, `replaced` and `loaded`, and within one answer the cache holds exactly the current pick, so `replaced` at a re-pick counts the experts that changed. Over the 135 re-picks in the 230-prompt set, our router's predicted churn tracks Apple's with $r = 0.934$ and a mean miss of 4.6 of about 40 changed experts. That bounds our picks to within a few of Apple's 1012 per pick. Moving the re-pick to a neighbouring token makes the match clearly worse (5.3–5.4), which pins the schedule. Every arithmetic variant scores the same within noise: 4.50 for the ANE-computed layer-12 state, 4.60 for the default, 4.60 for fp16 matmul outputs, 4.65 for fp32 with exact SiLU, 4.67 for exact SiLU, and 4.45 without the input bias at re-pick tokens. Apple's re-picks change slightly fewer experts than ours (by 1.6 on average). A small preference for already-loaded experts fits the counts only marginally better (4.60 to 4.40) and lowers agreement (95.15%), so we reject it. Apple's code runs a dense-only function at each re-pick token, the same procedure family as the prompt, which adds no layer-0 bias. Dropping the bias at those tokens raises agreement from 95.19% to 95.27% (12 fixed, 7 broken) and is part of the default. Between prompts the counts do not track the previous prompt's pick (Apple 501 on average, our prediction 111), so the cache is not simply carried over between requests.

Experiments on the uncorrected configuration. A further set of experiments was run before the layer-0 corrections (embedding $\times 100$, input bias on every token). With Apple's schedule, 74 of that configuration's 93 misses are also missed with per-token routing, so they do not depend on routing. At 89 of the 93 our choice is an ordinary text token, so Apple's output masking is not involved. Table 13 lists the numerical experiments.

Variant of the baseline (teacher-forced, 1127 tokens)	top-1
Baseline: Apple's schedule, int8 KV emulation, uncorrected layer 0	91.75%
int8 KV emulation off	90.77%
no re-pick during generation	90.95%
re-pick takes effect one token later	91.39%
activations rounded to fp16 (residual / + router input / + every sublayer output)	91.39%
RoPE angles computed in fp16 / + fp16 cos and sin	91.75% / 91.57%
attention scale $\times 0.8$ / $\times 0.9$ / $\times 1.1$ / $\times 1.25$ (of $1/\sqrt{128}$)	87.8 / 90.1 / 89.9 / 89.9%
tanh soft-cap of attention scores at 50 / 30 / 20 (Gemma-style)	91.4 / 90.4 / 89.5%
our own fp32 RoPE instead of llama.cpp's (same mathematics)	91.84%

Table 13. Numerical and schedule experiments on the uncorrected-layer-0 baseline. Emulating Apple's int8 KV cache and re-picking every 32 tokens are both worth close to a point, the attention scale peaks sharply at $1/\sqrt{128}$, and there is no attention soft-cap; storing activations or RoPE angles in fp16 changes nothing beyond the noise floor.

18. Responsible use

The model's weights belong to Apple. We publish only our own code (the llama.cpp architecture patch and the converter, at `https://github.com/yu314-coder/afm-ifp-teardown`), which rebuilds the model from the assets already on a user's Mac; we publish no weights, tokenizer data or decoded assets, and have asked Apple whether redistribution would be permitted. The project is unofficial and not affiliated with or endorsed by Apple.

Appendix A. GGUF layout

Tensor	Shape (rows × cols)	Type
`token_embd`	262144×1536	Q4_0
`input_bias`, `output_norm`, `output_norm_2`	1536	F32
`blk.L.attn_q`	2048×1536	Q2_K
`blk.L.attn_k`, `attn_v` ($L \leq 34$)	1024×1536	Q2_K
`blk.L.attn_output`	1536×2048	Q2_K
`blk.L.attn_q_norm`, `attn_k_norm` ($L < 34$)	128	F32
`blk.L.{post_attention,ffn,post_ffw,layer_output}_norm`	1536	F32
`blk.L.ffn_{gate,up}` / `ffn_down` ($L < 12$)	3072×1536 / 1536×3072	Q2_K
`blk.L.ffn_{gate,up}_shexp` / `ffn_down_shexp` ($L \geq 12$)	5888×1536 / 1536×5888	Q2_K
`blk.L.ffn_{gate,up}_exps` / `ffn_down_exps`	$348 \times 256 \times 1536$ / $348 \times 1536 \times 256$	Q2_K
`blk.12.ifp_router_fc1` (+bias), `_norm`, `_fc2` (+bias)	256×1536, 256, 15312×256	F32

Table 14. 805 tensors, 6.31 GiB. Metadata keys under `afm.*`: block count 56, embedding length 1536, head counts 16/8, key length 128, expert count 348, experts used 23, shared expert width 5888, leading dense blocks 12, KV-reuse first layer 35, RoPE base 5×10^5, RMS ε, embedding scale, per-layer K/V int8 scales, and a Jinja chat template.

Email address: `euler.yu@gmail.com`